

CAMBRIDGE INTERNATIONAL AS &amp; A LEVEL CHEMISTRY (9701)

# Topic 19 — Nitrogen Compounds (Amines & Nitriles)

Topical Past Paper Question Pack &amp; Comprehensive Mark Scheme

|                |                                                                                                                          |
|----------------|--------------------------------------------------------------------------------------------------------------------------|
| CANDIDATE      | Urwah                                                                                                                    |
| SYLLABUS       | Cambridge International AS Level Chemistry (9701)                                                                        |
| TOPIC COVERAGE | 19.1 Primary Amines · Basicity & Inductive Effects · Salt Formation & Acylation · 19.2 Nitriles · Hydrolysis & Reduction |
| EXAM SERIES    | Past Paper Compilation (2019 – 2025 Paper 2 Theory & Paper 1)                                                            |
| TOTAL MARKS    | 242 Marks (50 Questions)                                                                                                 |

## HOW TO USE THIS PACK

1. Complete each question carefully in the answer spaces provided, paying close attention to mark allocations.
2. Questions are classified into Easy and Hard tiers to guide progression from core recall to advanced data analysis.
3. Every question includes authentic Cambridge examination codes (e.g. 9701/22/M/J/22) for cross-referencing.
4. Full examiner marking points with point-by-point breakdowns are provided at the end of the document.

## SYLLABUS SUBTOPICS COVERED

### 19.1 Primary Amines & Basicity Trends

— Preparation from halogenoalkanes and excess ammonia; relative basicity (aliphatic primary amines)

### 19.1 Reactions of Primary Amines

— Salt formation with mineral acids ( $\text{RNH}_3^+\text{Cl}^-$ ) and liberation by strong base; acylation with acyl chlorides

### 19.2 Nitriles & Hydroxynitriles

— Nucleophilic substitution ( $\text{KCN}$ ) for +1 carbon chain extension; nucleophilic addition of  $\text{HCN}$  to carbonyl compounds

SUBTOPIC 19.1 — PRIMARY AMINES (STRUCTURE, BASICITY & REACTIONS)

1 Comparative Basicity of Nitrogen Compounds — 9701/22/M/J/23/Q5(a)–(d)

HARD · Syllabus 19.1

Amines behave as Bronsted-Lowry bases due to the presence of a lone pair of electrons on the nitrogen atom. Fig. 1.1 displays the comparative basicity and lone pair availability for ethylamine, ammonia, and phenylamine.

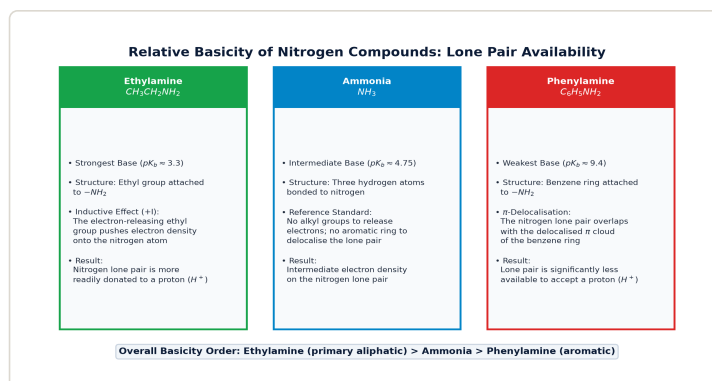


Fig. 1.1: Relative basicity and electron density of the nitrogen lone pair across aliphatic and aromatic nitrogen bases.

- ((a)) Define the term Bronsted-Lowry base, and write an equation showing how ethylamine acts as a base in aqueous solution. [2]

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- ((b)) Explain why ethylamine is a stronger base than ammonia, referring to inductive effects. [2]

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- ((c)) Explain why phenylamine (aniline) is a significantly weaker base than ammonia, referring to orbital overlap and electron delocalisation. [3]

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- ((d)) State the geometry and the approximate H-N-H bond angle in an ethylamine molecule, and compare it to the bond angle in the ethylammonium ion,  $\text{CH}_3\text{CH}_2\text{NH}_3^+$ . [3]

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2 Basicity Trends in Primary, Secondary & Tertiary Amines — 9701/21/O/N/23/Q4(a)–(c)

HARD · Syllabus 19.1

In aqueous solution, the basicity order of methylamines is: dimethylamine ( $(\text{CH}_3)_2\text{NH}$ ) > methylamine ( $\text{CH}_3\text{NH}_2$ ) > trimethylamine ( $(\text{CH}_3)_3\text{N}$ ) > ammonia ( $\text{NH}_3$ ).

- ((a)) Explain why dimethylamine is a stronger base than methylamine. [2]

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- ((b))** Explain why trimethylamine is a weaker base than dimethylamine in aqueous solution, despite having three electron-releasing methyl groups. Mention hydration and steric effects. **[3]**
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- ((c))** Predict whether diethylamine or ethylamine is the stronger base in water. **[1]**
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- .....

**3 Physical Properties and Hydrogen Bonding of Amines — 9701/22/F/M/22/Q3(a)–(c)**

EASY · Syllabus 19.1

Table 3.1 compares the boiling points and water solubilities of propane (b.p.  $-42^{\circ}\text{C}$ ), ethylamine (b.p.  $17^{\circ}\text{C}$ ), and ethanol (b.p.  $78^{\circ}\text{C}$ ), all having relative molecular masses  $M_r \approx 44$ – $46$ .

- ((a))** Explain why ethylamine has a significantly higher boiling point than propane. **[2]**
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- .....
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- ((b))** Explain why ethanol has a significantly higher boiling point than ethylamine, even though both substances exhibit intermolecular hydrogen bonding. **[2]**
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- .....
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- ((c))** Draw a labelled diagram showing hydrogen bonding between a molecule of ethylamine and a molecule of water. **[2]**
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**4 Salt Formation and Amine Liberation — 9701/23/M/J/23/Q4(a)–(c)**

EASY · Syllabus 19.1

Ethylamine dissolves in dilute hydrochloric acid to produce an odorless crystalline salt.

- ((a))** Write the balanced chemical equation for the reaction of ethylamine with hydrochloric acid. **[1]**
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- .....

- ((b))** Give the systematic name of the organic salt formed and describe the bonding present between the cation and anion. **[2]**
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- ((c))** Describe how the pungent, volatile ethylamine can be regenerated from this salt. Write an ionic equation for this reaction. **[2]**
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**5 Ligand Exchange and Complexation with Copper(II) — 9701/21/M/J/22/Q3(a)–(c)**

**HARD · Syllabus 19.1**

When aqueous butylamine is added dropwise to aqueous copper(II) sulfate, a pale blue precipitate appears. Upon adding excess butylamine, the precipitate dissolves to produce a deep royal-blue solution.

**((a))** Identify the chemical formula of the pale blue precipitate formed in the first step and state the property of butylamine that causes its precipitation. **[2]**

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**((b))** Identify the complex ion responsible for the deep royal-blue solution formed in excess butylamine, and state the type of bonding between butylamine and the copper ion. **[2]**

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**((c))** Write the ionic equation for the dissolution of the precipitate in excess butylamine (RNH<sub>2</sub>). **[2]**

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**6 Multiple Choice: Basicity Rankings — 9701/12/M/J/23/Q25**

**EASY · Syllabus 19.1**

Which compound is the strongest base in aqueous solution?

**((a))** Select the strongest base: **[1]**

- A Ammonia
- B Ethylamine
- C Phenylamine
- D Ethanamide

- A
- B
- C
- D

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**((b))** Explain why ethanamide is practically neutral (non-basic) in water. **[2]**

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**7 Reaction of Amines with Halogenoalkanes — 9701/22/O/N/22/Q3(a)–(d)**

**HARD · Syllabus 19.1**

Ethylamine reacts with bromoethane in a sealed tube under pressure.

**((a))** Write the equation for the reaction of ethylamine with bromoethane to form a secondary amine salt. **[1]**

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**((b))** Name the neutral secondary amine obtained after treatment with sodium hydroxide. **[1]**

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**((c))** Explain why this reaction does not stop at the secondary amine stage, leading to tertiary amines and quaternary ammonium salts. **[2]**

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**((d))** Give the formula and systematic name of the quaternary ammonium salt produced when excess bromoethane is used. **[2]**

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**8 Aromatic Amines: Synthesis of Phenylamine — 9701/21/O/N/21/Q4**

**HARD · Syllabus 19.1**

Phenylamine is manufactured from nitrobenzene via a two-stage reduction procedure.

**((a))** State the reagents and condition used in Stage 1 to reduce nitrobenzene to phenylammonium ions. **[2]**

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**((b))** State the reagent used in Stage 2 to liberate neutral phenylamine from the reaction mixture. **[1]**

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**((c))** Write the overall equation for the reduction of nitrobenzene to phenylamine using [H] to represent the reducing agent. **[1]**

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**9 Bromination of Phenylamine — 9701/23/O/N/22/Q5**

**HARD · Syllabus 19.1**

When aqueous bromine (bromine water) is added to phenylamine, an immediate reaction occurs without a catalyst.

**((a))** State two visible observations during this reaction. **[2]**

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**((b))** Draw the structural formula and give the systematic IUPAC name of the organic precipitate formed. **[2]**

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**((c))** Explain why phenylamine reacts vigorously with bromine water at room temperature without requiring a halogen carrier catalyst (such as FeBr<sub>3</sub> or AlBr<sub>3</sub>), unlike benzene. **[3]**

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**10 Acylation of Primary Amines to Form Amides — 9701/22/M/J/22/Q4**

**HARD · Syllabus 19.1**

Primary amines react vigorously with acyl chlorides at room temperature to form substituted amides.

**((a))** Write the balanced chemical equation for the reaction between ethanoyl chloride and ethylamine. **[1]**

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**((b))** Name the organic product formed and identify the functional group linking the two carbon chains. **[2]**

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**((c))** Explain why two molar equivalents of ethylamine are typically consumed for every mole of ethanoyl chloride in this reaction. **[2]**

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**11 Multiple Choice: Properties of Ethylamine — 9701/11/M/J/23/Q27**

**EASY · Syllabus 19.1**

Which statement regarding ethylamine is correct?

**((a))** Select the correct statement: **[1]**

- A It turns universal indicator paper red
- B It reacts with dilute hydrochloric acid to form an insoluble precipitate
- C It has a higher boiling point than ethanol
- D It reacts with ethanoyl chloride to produce N-ethylethanamide

A  
B  
C  
D

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**((b))** State the observation when ethylamine vapour comes into contact with damp red litmus paper. **[1]**

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**12 Diazotisation of Phenylamine (A Level Transition Link) — 9701/21/O/N/23/Q7**

**HARD · Syllabus 19.1**

When phenylamine is treated with nitrous acid ( $\text{HNO}_2$ ) at temperatures below  $5^\circ\text{C}$ , benzenediazonium chloride is formed.

**((a))** State how nitrous acid is generated in situ in the reaction mixture. **[1]**

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**((b))** Explain why the reaction temperature must be maintained strictly between  $0^\circ\text{C}$  and  $5^\circ\text{C}$ . **[2]**

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- ((c)) What happens if the benzenediazonium salt solution is warmed above 10 °C? State the observation and the organic product formed. [2]

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13 Distinguishing Aliphatic vs Aromatic Amines — 9701/22/F/M/21/Q4

EASY · Syllabus 19.1

Describe two distinct chemical tests to differentiate between liquid samples of butylamine and phenylamine.

- ((a)) Test 1: Describe the test using bromine water, stating observations for each compound. [2]

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- ((b)) Test 2: Describe the test using aqueous copper(II) sulfate, stating observations for each compound. [2]

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14 Synthetic Pathways to Primary Amines — 9701/21/M/J/23/Q8(a)–(d)

HARD · Syllabus 19.1

Primary aliphatic amines can be prepared either by direct nucleophilic substitution of halogenoalkanes with ammonia or by the reduction of nitriles. Fig. 14.1 compares these two synthetic routes.

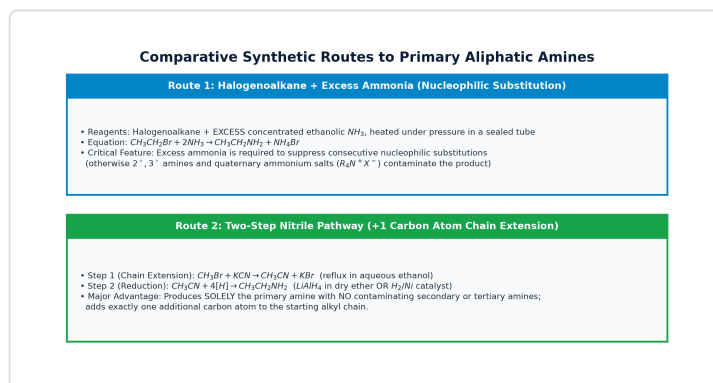


Fig. 14.1: Flowchart comparing Route 1 (halogenoalkane + excess ammonia) with Route 2 (nitrile extension and reduction).

- ((a)) In Route 1, bromoethane is heated with concentrated ammonia in a sealed tube. Explain why an excess of concentrated ammonia is essential. [2]

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- ((b)) Explain why the reaction mixture must be heated in a sealed glass tube rather than under an open reflux condenser. [2]

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- ((c)) In Route 2, bromomethane is converted into ethylamine. Identify the reagent in Step 1 and the reducing agent in Step 2. [2]

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((d)) State two synthetic advantages of Route 2 over Route 1 for preparing pure ethylamine. [2]

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**15 Consecutive Substitution Mechanism in Amine Preparation — 9701/22/O/N/23/Q5**

**HARD · Syllabus 19.1**

When 1-chloropropane reacts with ammonia, a mixture of four different organic products is obtained.

((a)) Write the balanced equations showing the successive steps from 1-chloropropane to propylamine, dipropylamine, and tripropylamine. [3]

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((b)) Give the formula and systematic IUPAC name of the fourth product, which is an ionic salt. [2]

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((c)) Explain why dipropylamine is a stronger nucleophile than propylamine. [2]

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**16 Separation of Amine Mixtures via Fractional Distillation — 9701/21/O/N/22/Q4**

**EASY · Syllabus 19.1**

Consider the boiling points of ethylamine (b.p. 17 °C), diethylamine (b.p. 55 °C), and triethylamine (b.p. 89 °C).

((a)) Explain why triethylamine has a higher boiling point than ethylamine, even though triethylamine CANNOT form hydrogen bonds between its own molecules. [3]

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((b)) State how the components of this amine mixture can be separated after liberating the free amines with aqueous sodium hydroxide. [1]

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**17 Synthesis of Amines from Amides: Reduction with LiAlH<sub>4</sub> — 9701/23/M/J/22/Q3**

**HARD · Syllabus 19.1**

Amides can be reduced to primary amines using lithium aluminium hydride in dry ether.

((a)) Write the balanced equation for the reduction of ethanamide, CH<sub>3</sub>CONH<sub>2</sub>, to ethylamine using [H] for the reducing agent. [1]

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**((b))** State the solvent used and explain why water must be strictly excluded until the reaction is complete. **[2]**

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**((c))** Contrast the product of reducing an amide with  $\text{LiAlH}_4$  to the product of reducing an amide by acid hydrolysis. **[2]**

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**18 Multiple Choice: Reagents for Amine Synthesis — 9701/12/F/M/23/Q25**

EASY · Syllabus 19.1

Which reaction converts 1-bromobutane into butylamine with no higher amine by-products?

**((a))** Select the correct two-step sequence: **[1]**

- A Step 1:  $\text{KCN}$  in ethanol; Step 2:  $\text{LiAlH}_4$  in dry ether
- B Step 1: Aqueous  $\text{NaOH}$ ; Step 2:  $\text{NH}_3$
- C Step 1:  $\text{Mg}$  in dry ether; Step 2:  $\text{NH}_3$
- D None of these
- A
- B
- C
- D

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**((b))** Explain why option A produces pentylamine rather than butylamine. **[2]**

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**19 Gabriel Phthalimide Synthesis (Curriculum Extension) — 9701/21/M/J/21/Q5**

HARD · Syllabus 19.1

The Gabriel synthesis is a laboratory method specifically developed to synthesise pure primary amines without forming secondary or tertiary amines.

**((a))** Explain why potassium phthalimide reacts with 1-bromobutane by nucleophilic substitution, but the resulting alkylated product CANNOT react with a second molecule of 1-bromobutane. **[2]**

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**((b))** State how the primary amine is liberated from the alkylphthalimide intermediate. **[1]**

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**20 Quaternary Ammonium Salts as Surfactants — 9701/22/M/J/23/Q10**

EASY · Syllabus 19.1

Quaternary ammonium salts containing long hydrocarbon chains, such as cetyltrimethylammonium bromide, are cationic surfactants used in fabric softeners and hair conditioners.

**((a))** Explain why quaternary ammonium ions carry a permanent positive charge regardless of the pH of the solution. **[2]**

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**((b))** Explain how cationic surfactants bind to negatively charged wet hair or cotton fabric fibres to reduce static cling and soften fibres. **[2]**

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**21 Chiral Amines: Optical Activity — 9701/23/O/N/23/Q4**

**HARD · Syllabus 19.1**

Butan-2-amine exhibits optical isomerism.

**((a))** Draw the structural formula of butan-2-amine and mark the chiral carbon atom with an asterisk (\*). **[1]**

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**((b))** Draw 3D wedge-and-dash representations showing the two non-superimposable optical enantiomers of butan-2-amine. **[2]**

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**((c))** Explain why both enantiomers exhibit identical chemical reactions with dilute hydrochloric acid. **[1]**

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**22 Multiple Choice: Mechanism of Amine Formation — 9701/11/O/N/22/Q25**

**EASY · Syllabus 19.1**

What type of reaction and mechanism is involved when bromoethane reacts with ammonia to form ethylamine?

**((a))** Select the correct reaction type and mechanism: **[1]**

A Electrophilic addition

B Nucleophilic addition

C Nucleophilic substitution (S<sub>N</sub>2)

D Free-radical substitution

A

B

C

D

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**((b))** Identify the nucleophile in this reaction. **[1]**

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**23 Diamines in Polyamide Manufacture — 9701/21/O/N/21/Q6**

HARD · Syllabus 19.1

Hexane-1,6-diamine,  $\text{H}_2\text{N}-(\text{CH}_2)_6-\text{NH}_2$ , is reacted with hexanedioic acid to manufacture Nylon 6,6.

- ((a))** [2]  
Write the balanced equation for the condensation polymerisation of hexane-1,6-diamine with hexanedioic acid, showing one repeat unit.

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- ((b))** [1]  
Identify the linkage that connects the monomer units in the nylon chain.

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- ((c))** [2]  
Explain why Nylon 6,6 fibres have high tensile strength and can be drawn into durable threads.

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**24 Deducing Amine Structure from Elemental Analysis & MS — 9701/22/F/M/23/Q3**

HARD · Syllabus 19.1

An organic primary amine D contains 65.7% carbon, 15.1% hydrogen, and 19.2% nitrogen by mass. Its mass spectrum shows a molecular ion at  $m/z = 73$ .

- ((a))** [2]  
Calculate the empirical formula and molecular formula of amine D.

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- ((b))** [4]  
Draw the structural formulas of the four isomeric primary amines with this molecular formula.

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- ((c))** [1]  
Which of these four isomers contains a chiral carbon atom?

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**25 Combustion of Amines and NO<sub>x</sub> Emissions — 9701/11/M/J/22/Q25**

EASY · Syllabus 19.1

When methylamine burns completely in oxygen, it produces carbon dioxide, water vapour, and nitrogen gas.

- ((a))** [2]  
Write a balanced chemical equation for the complete combustion of methylamine,  $\text{CH}_3\text{NH}_2$ .

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- ((b))** [1]  
In vehicle engines, incomplete combustion of nitrogenous fuels can produce nitrogen oxides ( $\text{NO}$  and  $\text{NO}_2$ ). State one environmental hazard caused by  $\text{NO}_x$  emissions.

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26 Multiple Choice: Basicity of Amine Derivatives — 9701/12/O/N/23/Q27

EASY · Syllabus 19.1

Which organic nitrogen compound does NOT form a salt when mixed with dilute sulfuric acid?

((a))

[1]

Select the correct compound:

- A Butylamine
  - B Diethylamine
  - C Ethanamide
  - D Phenylamine
- A
- B
- C
- D

((b))

[2]

Explain your reasoning.

SUBTOPIC 19.2 — NITRILES & HYDROXYNITRILES (SYNTHESIS, HYDROLYSIS & REDUCTION)

27 Comprehensive Chemical Reactions of Nitriles — 9701/21/M/J/23/Q9(a)-(d)

HARD · Syllabus 19.2

Nitriles ( $R-CN$ ) are vital synthetic intermediates that can be converted into either carboxylic acids or primary amines. Fig. 27.1 illustrates the dual hydrolysis pathways and reduction chemistry of propanenitrile.

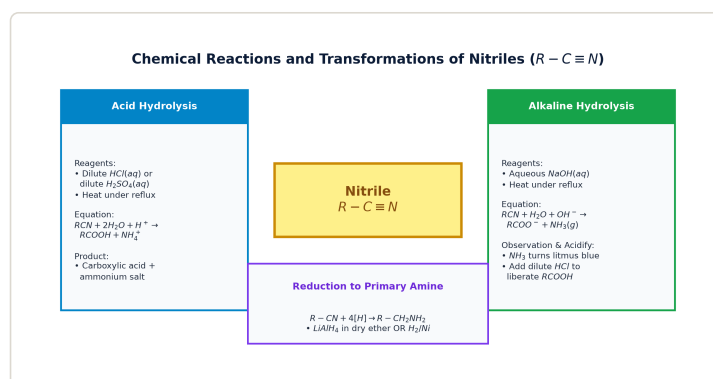


Fig. 27.1: Chemical transformation pathways of nitriles: acid hydrolysis, alkaline hydrolysis, and reduction.

((a))

[2]

State the reagents and conditions for the acid hydrolysis of propanenitrile, and write the balanced chemical equation.

((b))

[2]

State the reagents and conditions for the alkaline hydrolysis of propanenitrile, and write the balanced chemical equation.

((c))

[2]

State the observation during alkaline hydrolysis that confirms ammonia gas is evolved, and write the equation for liberating propanoic acid from the resulting solution.

- ((d)) State the reagent, solvent, and write the equation for the reduction of propanenitrile to propylamine using  $[H]$ . [2]
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**28 Synthesis of Nitriles from Halogenoalkanes — 9701/22/O/N/23/Q6(a)-(c)**

EASY · Syllabus 19.2

1-bromobutane is converted into pentanenitrile.

- ((a)) State the reagent and solvent required for this nucleophilic substitution reaction. [2]
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- .....

- ((b)) Write the balanced chemical equation. [1]
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- .....

- ((c)) Explain why potassium cyanide, KCN, is dissolved in ethanol/water mixture rather than pure water. [2]
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**29 2-Hydroxynitriles: Addition of HCN to Carbonyls — 9701/21/O/N/22/Q5**

HARD · Syllabus 19.2

Ethanal reacts with HCN in the presence of NaCN to form 2-hydroxypropanenitrile.

- ((a)) Outline the nucleophilic addition mechanism by drawing curly arrows showing the attack of  $:CN$  on ethanal and subsequent protonation. [3]
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- ((b)) State why this reaction is described as a 'step-up' reaction. [1]
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- ((c)) Write the equation for the acid hydrolysis of 2-hydroxypropanenitrile to lactic acid. [1]
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- .....

**30 Reduction of Nitriles:  $LiAlH_4$  vs Catalytic Hydrogenation — 9701/22/M/J/22/Q6**

HARD · Syllabus 19.2

Industrial manufacturing of hexamethylenediamine from adiponitrile uses catalytic hydrogenation, whereas laboratory preparations often use  $LiAlH_4$ .

- ((a)) Write the equation for the catalytic hydrogenation of butanenitrile,  $CH_3CH_2CH_2CN$ , to butan-1-amine using  $H_2$  and a nickel catalyst. [1]
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- .....

**((b))** State the conditions (temperature, pressure, catalyst) for industrial catalytic hydrogenation of nitriles. **[2]**

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**((c))** State two reasons why catalytic hydrogenation is preferred in chemical factories over using  $\text{LiAlH}_4$ . **[2]**

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**31 Nitrile Hydrolysis Kinetics and Mechanism — 9701/23/M/J/23/Q5**

**HARD · Syllabus 19.2**

During the acid hydrolysis of a nitrile, an amide is formed as an intermediate before full conversion to the carboxylic acid.

**((a))** Draw the structure of the amide intermediate formed during the partial hydrolysis of ethanenitrile,  $\text{CH}_3\text{CN}$ . **[1]**

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**((b))** Explain why ethanamide can be isolated if the reaction is carried out under mild conditions, but complete reflux with strong acid yields ethanoic acid. **[2]**

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**32 Multiple Choice: Reagents for Nitrile Reduction — 9701/11/M/J/23/Q28**

**EASY · Syllabus 19.2**

Which reagent does NOT reduce a nitrile to a primary amine?

**((a))** Select the reagent: **[1]**

- A  $\text{LiAlH}_4$  in dry ether
  - B  $\text{H}_2$  gas over a heated nickel catalyst
  - C  $\text{NaBH}_4$  in aqueous ethanol
  - D Sodium metal in boiling ethanol
- A
- B
- C
- D

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**((b))** Explain why your chosen reagent fails to reduce nitriles. **[2]**

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**33 Deducing Nitrile Structure from Hydrolysis Products — 9701/21/O/N/23/Q6**

**HARD · Syllabus 19.2**

A branched nitrile J with molecular formula  $\text{C}_5\text{H}_9\text{N}$  is hydrolysed under reflux with dilute sulfuric acid to produce carboxylic acid K ( $\text{C}_5\text{H}_{10}\text{O}_2$ ). Acid K displays three peaks in its carbon-13 NMR spectrum.

((a)) Deduce the structural formula and systematic IUPAC name of carboxylic acid K. [2]

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((b)) Deduce the structural formula and systematic name of nitrile J. [2]

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((c)) Draw the skeletal structure of the primary amine obtained when nitrile J is reduced with  $\text{LiAlH}_4$ . [1]

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**34 Multi-Step Synthesis: Alcohol to Amine (+1 Carbon) — 9701/22/M/J/22/Q7**

**HARD · Syllabus 19.2**

Devise a three-step synthetic pathway to convert ethanol into propylamine.

((a)) Step 1: Convert ethanol into bromoethane. State the reagents and conditions. [2]

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((b)) Step 2: Convert bromoethane into propanenitrile. State the reagents and conditions. [2]

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((c)) Step 3: Reduce propanenitrile to propylamine. State the reagents and conditions. [2]

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**35 Reaction of Cyanide with Halogenoarenes — 9701/21/O/N/21/Q5**

**HARD · Syllabus 19.2**

When chlorobenzene is heated with ethanolic KCN under reflux, NO reaction takes place, whereas 1-chlorobutane reacts rapidly.

((a)) Explain why halogenoarenes such as chlorobenzene are completely inert to nucleophilic substitution by cyanide ions under standard conditions. [3]

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((b)) Suggest how benzenecarbonitrile (benzonitrile,  $\text{C}_6\text{H}_5\text{CN}$ ) can be prepared from phenylamine instead. [2]

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**36 Hydrolysis Stoichiometry of Dinitriles — 9701/23/O/N/22/Q6**

**HARD · Syllabus 19.2**

Adiponitrile (hexanedinitrile),  $\text{NC}-(\text{CH}_2)_4-\text{CN}$ , is hydrolysed by boiling with excess dilute hydrochloric acid.

**((a))** Write the balanced equation for the complete acid hydrolysis of adiponitrile. **[2]**

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**((b))** Name the dicarboxylic acid produced. **[1]**

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**((c))** Calculate the mass of dicarboxylic acid formed from 5.40 g of adiponitrile ( $M_r = 108.1$ ), assuming an 85.0% yield. **[3]**

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**37 Multiple Choice: Chain Extension Identification — 9701/12/M/J/23/Q29**

**EASY · Syllabus 19.2**

Which reaction extends the carbon skeleton by ONE carbon atom?

**((a))** Select the correct reaction: **[1]**

- Select the correct reaction:
- A Bromoethane + ethanolic  $\text{NH}_3$
  - B Bromoethane + ethanolic  $\text{KCN}$
  - C Ethanol + acidified  $\text{K}_2\text{Cr}_2\text{O}_7$
  - D Ethanol +  $\text{PCl}_5$
- A
- B
- C
- D

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**((b))** Name the organic product formed in the reaction you selected. **[1]**

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**38 Reduction of Hydroxynitriles: Stereochemical Integrity — 9701/22/F/M/22/Q5**

**HARD · Syllabus 19.2**

Optically active (R)-2-hydroxypropanenitrile is reduced with  $\text{LiAlH}_4$  in dry ether to form 1-aminopropan-2-ol.

**((a))** Write the equation for this reduction using  $[\text{H}]$ . **[1]**

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**((b))** Explain why the chiral carbon centre retains its stereochemical configuration during this reaction. **[2]**

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**((c))** Draw the structural formula of 1-aminopropan-2-ol. **[1]**

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39 Nitriles vs Isonitriles (Carbylamines) — 9701/21/M/J/22/Q8

HARD · Syllabus 19.2

When bromoethane is reacted with KCN, propanenitrile is formed. However, when bromoethane is reacted with silver cyanide, AgCN, the major product is ethyl isonitrile (ethyl carbylamine, CH<sub>3</sub>CH<sub>2</sub>NC).

((a)) State the difference in structure and bonding between propanenitrile (CH<sub>3</sub>CH<sub>2</sub>CN) and ethyl isonitrile (CH<sub>3</sub>CH<sub>2</sub>NC). [2]

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((b)) Explain why AgCN favours attack through the nitrogen atom rather than the carbon atom, referring to covalent vs ionic character. [2]

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SUBTOPIC 19.1 — PRIMARY AMINES (STRUCTURE, BASICITY & REACTIONS)

40 Reactions of Amines: Acylation and Salt Formation — 9701/21/M/J/23/Q10(a)-(d)

HARD · Syllabus 19.1

Primary amines behave as both nucleophiles and bases. Fig. 40.1 illustrates three characteristic reaction profiles: salt formation with acids, acylation with acyl chlorides, and complexation with copper(II) ions.

**Key Reactions of Primary Amines: Salt Formation, Acylation & Complexation**

| Salt Formation with Acids                                                                                                                                                                                                                                                                                                                        | Acylation with Acyl Chlorides                                                                                                                                                                                                                                                                                                                                                      | Copper(II) Complexation                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Reaction with mineral acids:<br/> <math>RNH_2 + HCl \rightarrow RNH_3^+ Cl^-</math></p> <ul style="list-style-type: none"> <li>Reversible reaction</li> <li>Forms crystalline ionic salt</li> <li>Water-soluble ammonium salt</li> <li>Adding NaOH liberates free amine:<br/> <math>RNH_3^+ + OH^- \rightarrow RNH_2 + H_2O</math></li> </ul> | <p>Reaction with R'COCl:<br/> <math>R'COCl + RNH_2 \rightarrow R'CONHR + HCl</math></p> <ul style="list-style-type: none"> <li>Produces secondary substituted amide</li> <li>Forms peptide / amide link (<math>-CONH-</math>)</li> <li>Vigorous reaction at room temperature</li> <li>Second equivalent of amine reacts with HCl to form <math>RNH_3^+ Cl^-</math> salt</li> </ul> | <p>With aqueous copper(II) sulfate:</p> <ul style="list-style-type: none"> <li>Dropwise amine:<br/>                     Pale blue precipitate of <math>Cu(OH)_2</math> (amine acts as Brønsted base)</li> <li>Excess amine:<br/>                     Precipitate dissolves forming deep royal-blue solution of complex:<br/> <math>[Cu(RNH_2)_4(H_2O)_2]^{2+}</math></li> </ul> |

Fig. 40.1: Characteristic reactions of primary aliphatic amines: acid-base neutralisation, acylation, and coordination complexation.

((a)) Write the balanced chemical equation for the reaction between propylamine and propanoic acid, identifying the type of reaction. [2]

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((b)) Write the balanced chemical equation for the reaction of propylamine with propanoyl chloride at room temperature. [1]

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((c)) State the systematic IUPAC name of the substituted amide formed in (b). [1]

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- ((d)) Describe how you would distinguish between a solution of propylamine and a solution of propanamide using aqueous copper(II) sulfate. [3]

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41 Deducing Organic Nitrogen Compound G — 9701/22/O/N/23/Q7

HARD · Syllabus 19.1

An organic liquid G has molecular formula C<sub>3</sub>H<sub>9</sub>N.

- Liquid G turns damp red litmus paper blue.
- When G reacts with dilute hydrochloric acid, an ionic salt is formed.
- When G reacts with ethanoyl chloride, organic compound H (C<sub>5</sub>H<sub>11</sub>NO) is formed along with steamy white fumes.
- In the <sup>1</sup>H NMR spectrum, compound G displays only two signals with an integration ratio of 6:3.

- ((a)) State the functional group present in liquid G. [1]

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- ((b)) Use the <sup>1</sup>H NMR data to deduce the structural formula and systematic name of liquid G. [3]

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- ((c)) Draw the structural formula of compound H. [1]

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42 Infrared Spectroscopy of Nitrogen Compounds — 9701/21/O/N/22/Q7

HARD · Syllabus 19.1

The infrared spectra of primary amines, secondary amines, and nitriles display distinct characteristic absorptions.

- ((a)) State the characteristic wave number range for the CN triple bond stretching absorption in nitriles. [1]

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- ((b)) Primary amines display a doublet (two peaks) in the 3300–3500 cm<sup>-1</sup> region, whereas secondary amines display only a single peak. Explain this difference in terms of vibrational modes. [2]

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- ((c)) Explain why tertiary amines exhibit NO absorption bands in the 3300–3500 cm<sup>-1</sup> region. [1]

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**43 Mass Spectrometric Fragmentation of Amines (Alpha-Cleavage) — 9701/22/M/J/22/Q8**

**HARD · Syllabus 19.1**

The mass spectrum of propylamine,  $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$  ( $M_r = 59$ ), displays a base peak at  $m/z = 30$ .

- ((a))** Deduce the formula and structure of the resonance-stabilised iminium fragment ion responsible for the base peak at  $m/z = 30$ . **[2]**

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- ((b))** State what neutral radical is lost when the molecular ion fragments to form the peak at  $m/z = 30$ . **[1]**

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- ((c))** Explain why the base peak of propan-2-amine,  $\text{CH}_3\text{CH}(\text{NH}_2)\text{CH}_3$ , occurs at  $m/z = 44$  rather than  $m/z = 30$ . **[2]**

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**44 Multi-Step Synthesis: Alkene to Secondary Amide — 9701/23/M/J/23/Q7**

**HARD · Syllabus 19.1**

Devise a multi-step synthetic pathway to convert ethene into N-ethylpropanamide.

- ((a))** Outline the pathway to produce ethylamine from ethene in two steps. **[3]**

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- ((b))** Outline the pathway to produce propanoyl chloride from ethene in three steps. **[3]**

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- ((c))** Combine ethylamine and propanoyl chloride to produce N-ethylpropanamide, writing the equation and stating the conditions. **[2]**

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**SUBTOPIC 19.2 — NITRILES & HYDROXYNITRILES (SYNTHESIS, HYDROLYSIS & REDUCTION)**

**45 Multiple Choice: Nitrile Hydrolysis Outcomes — 9701/12/M/J/23/Q30**

**EASY · Syllabus 19.2**

Which compound is produced when 2-methylbutanenitrile is heated under reflux with dilute hydrochloric acid?

- ((a))** Select the correct IUPAC name: **[1]**

- A 2-methylbutan-1-amine  
B 2-methylbutanoic acid  
C 3-methylbutanoic acid  
D 2-methylbutanamide

- A  
B  
C  
D

**((b))**

Draw the skeletal formula of this product.

**[1]****SUBTOPIC 19.1 — PRIMARY AMINES (STRUCTURE, BASICITY & REACTIONS)****46****Comprehensive Chemical Roadmap: 3-Carbon Nitrogen Derivatives — 9701/21/O/N/23/Q9****HARD · Syllabus 19.1**

Consider the nitrogen-containing compounds: propanenitrile, propylamine, and propanamide.

**((a))**

State a chemical test that gives a positive result with propylamine but NO reaction with propanamide.

**[2]****((b))**

State the reagent and condition that converts propanenitrile into propylamine.

**[2]****((c))**

State the reagent and condition that converts propanenitrile into propanoic acid.

**[2]****47****Preparation of Amines from Carbonyls (Reductive Amination) — 9701/22/M/J/23/Q12****HARD · Syllabus 19.1**

In the pharmaceutical industry, amines are prepared by treating an aldehyde or ketone with ammonia in the presence of a reducing agent (reductive amination).

**((a))**When propanal reacts with ammonia, an imine intermediate ( $\text{CH}_3\text{CH}_2\text{CH}=\text{NH}$ ) is formed. Name the type of reaction that forms this imine.**[1]****((b))**When the imine is reduced with  $\text{H}_2$  and a nickel catalyst, propylamine is formed. Write the equation for this reduction.**[1]****((c))**

Deduce the structure of the amine formed when propanone is subjected to reductive amination with ammonia.

**[1]****48****Multiple Choice: Reactions of Phenylamine — 9701/11/O/N/23/Q26****EASY · Syllabus 19.1**

Which reagent reacts with phenylamine to produce a white precipitate?

**((a))**

Select the reagent:

A Aqueous sodium hydroxide

B Bromine water

C Dilute hydrochloric acid

D Ethanoyl chloride

A

B

C

D

**[1]**

((b)) Name the white precipitate.

[1]

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**49 Acid-Base Titration of Ethylamine — 9701/21/M/J/22/Q9**

**HARD · Syllabus 19.1**

A 20.0 cm<sup>3</sup> sample of an aqueous solution of ethylamine is titrated with 0.150 mol dm<sup>3</sup> hydrochloric acid. Exactly 18.40 cm<sup>3</sup> of HCl is required to reach the equivalence point.

((a)) Calculate the concentration of the ethylamine solution in mol dm<sup>3</sup>.

[2]

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((b)) State whether the pH at the equivalence point of this titration is acidic (pH < 7), neutral (pH = 7), or alkaline (pH > 7). Explain your answer in terms of the hydrolysis of the salt formed.

[2]

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((c)) Choose a suitable indicator for this titration from methyl orange (pH 3.1–4.4) and phenolphthalein (pH 8.3–10.0). Justify your choice.

[2]

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**50 Comprehensive Deduction of Unknown Nitrogen Compound Z — 9701/22/M/J/23/Q13**

**HARD · Syllabus 19.1**

Compound Z has molecular formula C<sub>4</sub>H<sub>9</sub>NO.

- When Z is heated under reflux with dilute sodium hydroxide, an alkaline gas is evolved that turns damp red litmus paper blue, and a sodium carboxylate salt is formed.
- When the carboxylate salt is acidified, propanoic acid is obtained.
- When Z is reduced with LiAlH<sub>4</sub> in dry ether, compound W (C<sub>4</sub>H<sub>11</sub>N) is formed.
- Compound W is a primary amine.

((a)) Identify the alkaline gas evolved during alkaline hydrolysis.

[1]

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((b)) Deduce the structural formula and systematic name of compound Z.

[3]

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((c)) Deduce the structural formula and systematic name of primary amine W.

[2]

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## Mark Scheme — Topic 19 — Nitrogen Compounds (Amines & Nitriles)

CAMBRIDGE INTERNATIONAL AS LEVEL CHEMISTRY (9701) — THEORY MARKING POINTS

| Q    | EXAMINER MARKING POINTS / INDICATIVE CONTENT                                                                                                                                                                                                                                                                                                                                                                                                           | MARKS |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1(a) | A proton (H) acceptor [1] $\text{CH}_3\text{CH}_2\text{NH}_2 + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{CH}_2\text{NH}_3^+ + \text{OH}^-$ [1]                                                                                                                                                                                                                                                                                            | 2     |
| 1(b) | The ethyl group exerts a positive inductive (+I) electron-releasing effect [1] This increases the electron density on the nitrogen atom, making the lone pair more readily available to accept a proton (H) [1]                                                                                                                                                                                                                                        | 2     |
| 1(c) | The lone pair of electrons on the nitrogen p-orbital overlaps with the delocalised pi electron system of the benzene ring [1] The lone pair becomes delocalised into the aromatic ring, significantly decreasing electron density on nitrogen [1] Consequently, the lone pair is far less available to coordinate with / accept a proton [1]                                                                                                           | 3     |
| 1(d) | In ethylamine: Trigonal pyramidal geometry with bond angle $\sim 107^\circ$ (due to lone pair–bonding pair repulsion) [1] In ethylammonium ion: Tetrahedral geometry around nitrogen with bond angle $\sim 109.5^\circ$ [1] Loss of lone pair into a dative bond eliminates lone pair repulsion, widening bond angles [1]                                                                                                                              | 3     |
| 2(a) | Dimethylamine has two electron-releasing methyl groups (+I effect) compared to one in methylamine [1] This results in greater electron density on the nitrogen lone pair, making it more attractive to protons [1]                                                                                                                                                                                                                                     | 2     |
| 2(b) | Steric hindrance: Three bulky methyl groups crowd the nitrogen atom, impeding the approach of H and water molecules [1] Hydration energy: The trimethylammonium ion $((\text{CH}_3)_3\text{NH}^+)$ has only one N–H bond to form hydrogen bonds with solvent water molecules, whereas $(\text{CH}_3)_2\text{NH}_2$ has two N–H bonds [1] Less favourable solvation / hydration energy destabilises the conjugate cation of trimethylamine in water [1] | 3     |
| 2(c) | Diethylamine $((\text{C}_2\text{H}_5)_2\text{NH})$ is the stronger base [1]                                                                                                                                                                                                                                                                                                                                                                            | 1     |
| 3(a) | Propane is non-polar and held only by weak London dispersion forces [1] Ethylamine has polar N–H bonds and forms intermolecular hydrogen bonds between molecules, requiring more energy to separate [1]                                                                                                                                                                                                                                                | 2     |
| 3(b) | Oxygen is more electronegative than nitrogen (electronegativity 3.5 vs 3.0) [1] The O–H bond is more polarised than the N–H bond, resulting in stronger intermolecular hydrogen bonds in ethanol than in ethylamine [1]                                                                                                                                                                                                                                | 2     |
| 3(c) | Correct dipoles: $\text{N}(\delta^-) - \text{H}(\delta^+)$ on ethylamine and $\text{H}(\delta^+) - \text{O}(\delta^-)$ on water [1] Dashed line from lone pair on nitrogen to $\text{H}(\delta^+)$ of water (or from water oxygen lone pair to $\text{H}(\delta^+)$ of amine) [1]                                                                                                                                                                      | 2     |
| 4(a) | $\text{CH}_3\text{CH}_2\text{NH}_2 + \text{HCl} \rightarrow \text{CH}_3\text{CH}_2\text{NH}_3^+ \text{Cl}^-$ [1]                                                                                                                                                                                                                                                                                                                                       | 1     |
| 4(b) | Ethylammonium chloride [1] Ionic bonding / strong electrostatic attraction between positive $\text{CH}_3\text{CH}_2\text{NH}_3^+$ and negative $\text{Cl}^-$ ions [1]                                                                                                                                                                                                                                                                                  | 2     |
| 4(c) | Add a strong alkali such as aqueous sodium hydroxide $(\text{NaOH(aq)})$ and warm gently [1] $\text{CH}_3\text{CH}_2\text{NH}_3^+ + \text{OH}^- \rightarrow \text{CH}_3\text{CH}_2\text{NH}_2 + \text{H}_2\text{O}$ [1]                                                                                                                                                                                                                                | 2     |
| 5(a) | Precipitate formula: $\text{Cu}(\text{OH})_2$ / copper(II) hydroxide [1] Butylamine acts as a Bronsted–Lowry base in water, generating $\text{OH}^-$ ions $(\text{RNH}_2 + \text{H}_2\text{O} \rightleftharpoons \text{RNH}_3^+ + \text{OH}^-)$ which precipitate $\text{Cu}^{2+}$ [1]                                                                                                                                                                 | 2     |
| 5(b) | $[\text{Cu}(\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2)_4(\text{H}_2\text{O})_2]^{2+}$ (or $[\text{Cu}(\text{RNH}_2)_4]^{2+}$ ) [1] Dative covalent (coordinate) bonding via the nitrogen lone pair [1]                                                                                                                                                                                                                                   | 2     |
| 5(c) | $\text{Cu}(\text{OH})_2(\text{s}) + 4\text{RNH}_2(\text{aq}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow [\text{Cu}(\text{RNH}_2)_4(\text{H}_2\text{O})_2]^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq})$ [2]                                                                                                                                                                                                                                              | 2     |
| 6(a) | B (Ethylamine) [1]                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1     |
| 6(b) | In ethanamide $(\text{CH}_3\text{CONH}_2)$ , the lone pair of electrons on nitrogen is adjacent to an electronegative carbonyl group $(\text{C}=\text{O})$ [1] The nitrogen lone pair is strongly delocalised into the pi system of the carbonyl group, rendering it completely unavailable for protonation [1]                                                                                                                                        | 2     |
| 7(a) | $\text{CH}_3\text{CH}_2\text{NH}_2 + \text{CH}_3\text{CH}_2\text{Br} \rightarrow (\text{CH}_3\text{CH}_2)_2\text{NH}_2\text{Br}$ [1]                                                                                                                                                                                                                                                                                                                   | 1     |
| 7(b) | Diethylamine [1]                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1     |

| Q     | EXAMINER MARKING POINTS / INDICATIVE CONTENT                                                                                                                                                                                                                                                                                                                                                                                                                                                     | MARKS |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 7(c)  | The secondary amine formed retains a lone pair on nitrogen and is an even stronger nucleophile than the starting primary amine due to two electron-releasing ethyl groups [1] It competes effectively for remaining bromoethane molecules, undergoing successive nucleophilic substitutions [1]                                                                                                                                                                                                  | 2     |
| 7(d)  | Formula: $(\text{CH}_3\text{CH}_2)_4\text{NBr}$ [1] Name: Tetraethylammonium bromide [1]                                                                                                                                                                                                                                                                                                                                                                                                         | 2     |
| 8(a)  | Tin (Sn) and concentrated hydrochloric acid (conc. HCl) [1] Heat under reflux (or boiling water bath) [1]                                                                                                                                                                                                                                                                                                                                                                                        | 2     |
| 8(b)  | Aqueous sodium hydroxide ( $\text{NaOH(aq)}$ ) in excess [1]                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1     |
| 8(c)  | $\text{C}_6\text{H}_5\text{NO}_2 + 6[\text{H}] \rightarrow \text{C}_6\text{H}_5\text{NH}_2 + 2\text{H}_2\text{O}$ [1]                                                                                                                                                                                                                                                                                                                                                                            | 1     |
| 9(a)  | Bromine water is decolourised (orange-brown to colourless) [1] White precipitate forms immediately [1]                                                                                                                                                                                                                                                                                                                                                                                           | 2     |
| 9(b)  | Structure showing 2,4,6-tribromophenylamine (benzene ring with $\text{-NH}_2$ at position 1 and Br atoms at 2, 4, 6) [1] Systematic name: 2,4,6-tribromophenylamine [1]                                                                                                                                                                                                                                                                                                                          | 2     |
| 9(c)  | The lone pair of electrons on the nitrogen atom is delocalised into the benzene ring's pi electron system [1] This significantly activates the ring by increasing the pi electron density (particularly at 2, 4, and 6 positions) [1] The enhanced electron density polarises incoming bromine molecules effectively without needing a Lewis acid catalyst [1]                                                                                                                                   | 3     |
| 10(a) | $\text{CH}_3\text{COCl} + \text{CH}_3\text{CH}_2\text{NH}_2 \rightarrow \text{CH}_3\text{CONHCH}_2\text{CH}_3 + \text{HCl}$ [1]                                                                                                                                                                                                                                                                                                                                                                  | 1     |
| 10(b) | N-ethylethanamide [1] Secondary amide (or peptide) linkage: $\text{-CONH-}$ [1]                                                                                                                                                                                                                                                                                                                                                                                                                  | 2     |
| 10(c) | The reaction liberates acidic hydrogen chloride gas (HCl) [1] The basic ethylamine acts as a scavenger, reacting with HCl to form ethylammonium chloride salt ( $\text{CH}_3\text{CH}_2\text{NH}_3\text{Cl}$ ), consuming a second equivalent of amine [1]                                                                                                                                                                                                                                       | 2     |
| 11(a) | D (It reacts with ethanoyl chloride to produce N-ethylethanamide) [1]                                                                                                                                                                                                                                                                                                                                                                                                                            | 1     |
| 11(b) | Damp red litmus paper turns blue (alkaline gas) [1]                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1     |
| 12(a) | Sodium nitrite ( $\text{NaNO}_2$ ) and excess dilute hydrochloric acid (HCl) [1]                                                                                                                                                                                                                                                                                                                                                                                                                 | 1     |
| 12(b) | The diazonium ion ( $\text{C}_6\text{H}_5\text{N}_2^+$ ) is unstable and decomposes rapidly at higher temperatures [1] Low temperature ( $< 5^\circ\text{C}$ ) prevents thermal decomposition while remaining warm enough for reaction to occur [1]                                                                                                                                                                                                                                              | 2     |
| 12(c) | Effervescence / bubbling of nitrogen gas ( $\text{N}_2$ ) [1] Organic product: Phenol ( $\text{C}_6\text{H}_5\text{OH}$ ) [1]                                                                                                                                                                                                                                                                                                                                                                    | 2     |
| 13(a) | Butylamine: No precipitate, bromine water remains orange-brown (no reaction) [1] Phenylamine: Bromine water decolourises immediately, forming a white precipitate of 2,4,6-tribromophenylamine [1]                                                                                                                                                                                                                                                                                               | 2     |
| 13(b) | Butylamine: Pale blue precipitate formed which dissolves in excess to give a deep royal-blue solution [1] Phenylamine: Insoluble / green-brown precipitate or no deep blue complex (too weakly basic and insoluble in water) [1]                                                                                                                                                                                                                                                                 | 2     |
| 14(a) | Excess ammonia ensures that ammonia molecules outcompete the formed ethylamine for unreacted bromoethane [1] This suppresses consecutive nucleophilic substitutions that would otherwise yield secondary, tertiary amines, and quaternary ammonium salts [1]                                                                                                                                                                                                                                     | 2     |
| 14(b) | Ammonia is an extremely volatile gas (b.p. $-33^\circ\text{C}$ ) and would escape from an open reflux condenser [1] A sealed tube maintains high pressure, keeping ammonia in concentrated solution and driving the reaction forward [1]                                                                                                                                                                                                                                                         | 2     |
| 14(c) | Step 1 reagent: Potassium cyanide (KCN) or sodium cyanide in aqueous ethanol, heat under reflux [1] Step 2 reducing agent: Lithium aluminium hydride ( $\text{LiAlH}_4$ ) in dry ether OR $\text{H}_2$ with Ni catalyst [1]                                                                                                                                                                                                                                                                      | 2     |
| 14(d) | Route 2 produces exclusively primary amine without any secondary or tertiary amine contaminants [1] Route 2 extends the carbon chain by exactly one carbon atom (+1 C), which is valuable in step-up organic synthesis [1]                                                                                                                                                                                                                                                                       | 2     |
| 15(a) | $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + 2\text{NH}_3 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2 + \text{NH}_4\text{Cl}$ [1] $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + \text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2 \rightarrow (\text{CH}_3\text{CH}_2\text{CH}_2)_2\text{NH} + \text{HCl}$ [1] $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + (\text{CH}_3\text{CH}_2\text{CH}_2)_2\text{NH} \rightarrow (\text{CH}_3\text{CH}_2\text{CH}_2)_3\text{N} + \text{HCl}$ [1] | 3     |

| Q     | EXAMINER MARKING POINTS / INDICATIVE CONTENT                                                                                                                                                                                                                                                                                                                                                         | MARKS |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 15(b) | Formula: $(\text{CH}_3\text{CH}_2\text{CH}_2)_4\text{NCl}$ [1] Name: Tetrapropylammonium chloride [1]                                                                                                                                                                                                                                                                                                | 2     |
| 15(c) | Dipropylamine has two electron-releasing propyl groups (+I effect) attached to nitrogen compared to one in propylamine [1] This increases the electron density on the nitrogen lone pair, making it attack the $\delta^+$ carbon of the halogenoalkane faster [1]                                                                                                                                    | 2     |
| 16(a) | Triethylamine ( $M_r = 101.2$ ) has far more electrons and a much larger molecular surface area than ethylamine ( $M_r = 45.1$ ) [1] It possesses significantly stronger London dispersion (induced dipole-dipole) forces between its molecules [1] The cumulative dispersion forces in triethylamine outweigh the hydrogen bonding in smaller ethylamine, requiring more thermal energy to boil [1] | 3     |
| 16(b) | Fractional distillation [1]                                                                                                                                                                                                                                                                                                                                                                          | 1     |
| 17(a) | $\text{CH}_3\text{CONH}_2 + 4[\text{H}] \rightarrow \text{CH}_3\text{CH}_2\text{NH}_2 + \text{H}_2\text{O}$ [1]                                                                                                                                                                                                                                                                                      | 1     |
| 17(b) | Dry ether (ethoxyethane) [1] $\text{LiAlH}_4$ reacts violently with water to produce flammable hydrogen gas and hydrolyses the reagent [1]                                                                                                                                                                                                                                                           | 2     |
| 17(c) | Reduction with $\text{LiAlH}_4$ yields an amine (preserves nitrogen in organic chain, converts $\text{C}=\text{O}$ to $-\text{CH}_2-$ ) [1] Acid hydrolysis cleaves the $\text{C}-\text{N}$ bond, yielding a carboxylic acid and an ammonium salt [1]                                                                                                                                                | 2     |
| 18(a) | D (None of these) [1]                                                                                                                                                                                                                                                                                                                                                                                | 1     |
| 18(b) | Option A adds a nitrile carbon ( $\text{KCN}$ ), converting 4-carbon 1-bromobutane into 5-carbon pentanenitrile [1] Subsequent reduction yields pentan-1-amine (5 carbons), NOT butylamine (4 carbons) [1]                                                                                                                                                                                           | 2     |
| 19(a) | In N-butylphthalimide, the nitrogen lone pair is delocalised into two adjacent carbonyl groups [1] This makes nitrogen non-nucleophilic, completely preventing any further alkylation / consecutive substitution [1]                                                                                                                                                                                 | 2     |
| 19(b) | Alkaline hydrolysis with aqueous sodium hydroxide (or hydrazinolysis with hydrazine, $\text{NH}_2\text{NH}_2$ ) [1]                                                                                                                                                                                                                                                                                  | 1     |
| 20(a) | The nitrogen atom is bonded to four carbon atoms and possesses no lone pair or ionisable hydrogen atom [1] It cannot lose or gain a proton, so its formal +1 charge is permanent and independent of pH [1]                                                                                                                                                                                           | 2     |
| 20(b) | The positively charged quaternary head groups bond electrostatically to the negative charges on fibre surfaces [1] The non-polar hydrocarbon tails face outwards, providing a smooth, lubricated hydrophobic coating that reduces friction and static buildup [1]                                                                                                                                    | 2     |
| 21(a) | $\text{CH}_3-\text{C}^*(\text{NH}_2)-\text{CH}_2-\text{CH}_3$ with asterisk on C2 [1]                                                                                                                                                                                                                                                                                                                | 1     |
| 21(b) | Correct 3D tetrahedral representations using wedges, dashed lines, and solid bonds [1] Both enantiomers drawn as exact mirror images [1]                                                                                                                                                                                                                                                             | 2     |
| 21(c) | Hydrochloric acid is an achiral reagent; enantiomers possess identical chemical reactivity with all achiral substances [1]                                                                                                                                                                                                                                                                           | 1     |
| 22(a) | C (Nucleophilic substitution ( $\text{S}_\text{N}2$ )) [1]                                                                                                                                                                                                                                                                                                                                           | 1     |
| 22(b) | Ammonia molecule ( $:\text{NH}_3$ , via its lone pair of electrons on nitrogen) [1]                                                                                                                                                                                                                                                                                                                  | 1     |
| 23(a) | $-\text{[HN}-(\text{CH}_2)_6\text{-NH-CO}-(\text{CH}_2)_4\text{-CO]}- + 2\text{nH}_2\text{O}$ (or shown as repeat unit with continuation bonds) [2]                                                                                                                                                                                                                                                  | 2     |
| 23(b) | Amide link / peptide link ( $-\text{CO}-\text{NH}-$ ) [1]                                                                                                                                                                                                                                                                                                                                            | 1     |
| 23(c) | Linear polymer chains align parallel to one another [1] Forms extensive intermolecular hydrogen bonds between $\text{C}=\text{O}$ of one chain and $\text{N}-\text{H}$ of adjacent chains, providing high tensile strength [1]                                                                                                                                                                       | 2     |
| 24(a) | Moles: $\text{C} = 65.7/12 = 5.475$ ; $\text{H} = 15.1/1.0 = 15.1$ ; $\text{N} = 19.2/14 = 1.371$ [1] Ratio $\text{C}:\text{H}:\text{N} = 4:11:1 \rightarrow$ Empirical and Molecular formula = $\text{C}_4\text{H}_{11}\text{N}$ ( $M_r = 73$ ) [1]                                                                                                                                                 | 2     |
| 24(b) | Butan-1-amine: $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$ [1] Butan-2-amine: $\text{CH}_3\text{CH}(\text{NH}_2)\text{CH}_2\text{CH}_3$ [1] 2-methylpropan-1-amine: $(\text{CH}_3)_2\text{CHCH}_2\text{NH}_2$ [1] 2-methylpropan-2-amine: $(\text{CH}_3)_3\text{CNH}_2$ [1]                                                                                                            | 4     |

| Q     | EXAMINER MARKING POINTS / INDICATIVE CONTENT                                                                                                                                                                                                                                                                                                                                                          | MARKS |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 24(c) | Butan-2-amine [1]                                                                                                                                                                                                                                                                                                                                                                                     | 1     |
| 25(a) | $4\text{CH}_3\text{NH}_2 + 9\text{O}_2 \rightarrow 4\text{CO}_2 + 10\text{H}_2\text{O} + 2\text{N}_2$ [2]                                                                                                                                                                                                                                                                                             | 2     |
| 25(b) | Contributes to acid rain ( $\text{HNO}_3$ formation) / formation of photochemical smog / respiratory irritation [1]                                                                                                                                                                                                                                                                                   | 1     |
| 26(a) | C (Ethanamide) [1]                                                                                                                                                                                                                                                                                                                                                                                    | 1     |
| 26(b) | Ethanamide is an amide, not an amine [1]<br>The lone pair of electrons on nitrogen is delocalised into the adjacent carbonyl $\text{C}=\text{O}$ group, making it neutral and non-basic; it cannot be protonated by dilute mineral acid [1]                                                                                                                                                           | 2     |
| 27(a) | Reagents/conditions: Dilute hydrochloric acid (or dilute $\text{H}_2\text{SO}_4$ ), heat under reflux [1]<br>$\text{CH}_3\text{CH}_2\text{CN} + 2\text{H}_2\text{O} + \text{HCl} \rightarrow \text{CH}_3\text{CH}_2\text{COOH} + \text{NH}_4\text{Cl}$ (or $+\text{H} \rightarrow \dots + \text{NH}_4$ ) [1]                                                                                          | 2     |
| 27(b) | Reagents/conditions: Aqueous sodium hydroxide ( $\text{NaOH(aq)}$ ), heat under reflux [1]<br>$\text{CH}_3\text{CH}_2\text{CN} + \text{H}_2\text{O} + \text{NaOH} \rightarrow \text{CH}_3\text{CH}_2\text{COONa} + \text{NH}_3$ [1]                                                                                                                                                                   | 2     |
| 27(c) | Pungent gas evolved turns damp red litmus paper blue [1]<br>Add dilute strong acid: $\text{CH}_3\text{CH}_2\text{COO} + \text{H} \rightarrow \text{CH}_3\text{CH}_2\text{COOH}$ [1]                                                                                                                                                                                                                   | 2     |
| 27(d) | Reagent/solvent: $\text{LiAlH}_4$ in dry ether (or $\text{H}_2$ / Ni catalyst) [1]<br>$\text{CH}_3\text{CH}_2\text{CN} + 4[\text{H}] \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ [1]                                                                                                                                                                                                    | 2     |
| 28(a) | Reagent: Potassium cyanide ( $\text{KCN}$ ) or sodium cyanide ( $\text{NaCN}$ ) [1]<br>Solvent: Ethanol / aqueous ethanol, heat under reflux [1]                                                                                                                                                                                                                                                      | 2     |
| 28(b) | $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br} + \text{KCN} \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CN} + \text{KBr}$ [1]                                                                                                                                                                                                                                               | 1     |
| 28(c) | 1-bromobutane is insoluble in water (two immiscible layers would form, giving very slow reaction) [1]<br>Ethanol acts as a co-solvent that dissolves both the non-polar halogenoalkane and the ionic cyanide salt, creating a homogeneous reaction mixture [1]                                                                                                                                        | 2     |
| 29(a) | Curly arrow from lone pair on carbon of $:\text{CN}$ to $\text{C}(\delta+)$ of $\text{CH}_3\text{CHO}$ [1]<br>Curly arrow from $\text{C}=\text{O}$ double bond onto oxygen forming intermediate $\text{CH}_3\text{CH}(\text{CN})\text{O}^-$ [1]<br>Curly arrow from lone pair on $\text{O}^-$ to $\text{H}$ of $\text{HCN}$ (or $\text{H}_2\text{O}$ ) regenerating $:\text{CN}$ [1]                  | 3     |
| 29(b) | It extends the length of the carbon chain by adding one carbon atom (from 2 carbons in ethanal to 3 carbons in the product) [1]                                                                                                                                                                                                                                                                       | 1     |
| 29(c) | $\text{CH}_3\text{CH}(\text{OH})\text{CN} + 2\text{H}_2\text{O} + \text{H} \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{COOH} + \text{NH}_4^+$ [1]                                                                                                                                                                                                                                                | 1     |
| 30(a) | $\text{CH}_3\text{CH}_2\text{CH}_2\text{CN} + 2\text{H}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$ [1]                                                                                                                                                                                                                                                                    | 1     |
| 30(b) | Nickel (Ni) or platinum (Pt) catalyst [1]<br>Moderately high temperature ( $\sim 150\text{--}200^\circ\text{C}$ ) and elevated pressure ( $\sim 10\text{--}50$ atm) [1]                                                                                                                                                                                                                               | 2     |
| 30(c) | $\text{H}_2$ gas is much cheaper and readily available in bulk compared to expensive $\text{LiAlH}_4$ [1]<br>Catalytic hydrogenation has 100% atom economy with zero stoichiometric metal waste ( $\text{LiAlH}_4$ produces large amounts of aluminium hydroxide waste) [1]                                                                                                                           | 2     |
| 31(a) | $\text{CH}_3\text{CONH}_2$ (ethanamide) [1]                                                                                                                                                                                                                                                                                                                                                           | 1     |
| 31(b) | Addition of one water molecule across the $\text{CN}$ triple bond forms the amide: $\text{CH}_3\text{CN} + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{CONH}_2$ [1]<br>Prolonged heating with strong acid supplies the activation energy needed to hydrolyse the amide linkage: $\text{CH}_3\text{CONH}_2 + \text{H}_2\text{O} + \text{H} \rightarrow \text{CH}_3\text{COOH} + \text{NH}_4^+$ [1] | 2     |
| 32(a) | C ( $\text{NaBH}_4$ in aqueous ethanol) [1]                                                                                                                                                                                                                                                                                                                                                           | 1     |
| 32(b) | $\text{NaBH}_4$ is a relatively mild reducing agent that only reduces aldehydes and ketones [1]<br>It lacks sufficient hydride transfer power to reduce the less electrophilic, highly stable $\text{CN}$ triple bond [1]                                                                                                                                                                             | 2     |
| 33(a) | Acid K has 3 carbon environments: 2,2-dimethylpropanoic acid, $(\text{CH}_3)_3\text{C}-\text{COOH}$ [1]<br>Systematic name: 2,2-dimethylpropanoic acid [1]                                                                                                                                                                                                                                            | 2     |

| Q     | EXAMINER MARKING POINTS / INDICATIVE CONTENT                                                                                                                                                                                                                                                                                                                                                            | MARKS |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 33(b) | Structural formula: $(\text{CH}_3)_3\text{C}-\text{CN}$ [1] <br/> Systematic name: 2,2-dimethylpropanenitrile [1]                                                                                                                                                                                                                                                                                       | 2     |
| 33(c) | Skeletal formula of 2,2-dimethylpropan-1-amine: $(\text{CH}_3)_3\text{C}-\text{CH}_2-\text{NH}_2$ [1]                                                                                                                                                                                                                                                                                                   | 1     |
| 34(a) | Reagents: Sodium bromide (NaBr) and concentrated sulfuric acid (conc. $\text{H}_2\text{SO}_4$ ) (or $\text{PBr}_3$ / red P + $\text{Br}_2$ ) [1] <br/> Condition: Heat under reflux [1]                                                                                                                                                                                                                 | 2     |
| 34(b) | Reagent: Potassium cyanide (KCN) or NaCN [1] <br/> Condition: Aqueous ethanol, heat under reflux [1]                                                                                                                                                                                                                                                                                                    | 2     |
| 34(c) | Reagent: Lithium aluminium hydride ( $\text{LiAlH}_4$ ) in dry ether (or $\text{H}_2$ with heated Ni catalyst) [2]                                                                                                                                                                                                                                                                                      | 2     |
| 35(a) | The lone pair of electrons on chlorine p-orbital overlaps with the delocalised pi electron cloud of the benzene ring [1] <br/> This gives the C-Cl bond partial double bond character, making it significantly stronger and shorter than an aliphatic C-Cl bond [1] <br/> The electron-rich pi cloud of the benzene ring repels incoming nucleophiles ( $:\text{CN}$ ) preventing backside approach [1] | 3     |
| 35(b) | Diazotise phenylamine with $\text{NaNO}_2$ + HCl at $< 5^\circ\text{C}$ to form benzenediazonium chloride [1] <br/> React with copper(I) cyanide ( $\text{CuCN}$ ) / Sandmeyer reaction [1]                                                                                                                                                                                                             | 2     |
| 36(a) | $\text{NC}(\text{CH}_2)_4\text{CN} + 4\text{H}_2\text{O} + 2\text{HCl} \rightarrow \text{HOOC}(\text{CH}_2)_4\text{COOH} + 2\text{NH}_4\text{Cl}$ [2]                                                                                                                                                                                                                                                   | 2     |
| 36(b) | Hexanedioic acid (adipic acid) [1]                                                                                                                                                                                                                                                                                                                                                                      | 1     |
| 36(c) | Moles of adiponitrile = $5.40 / 108.1 = 0.04995 \text{ mol}$ [1] <br/> Theoretical mass of hexanedioic acid ( $M_r = 146.1$ ) = $0.04995 \times 146.1 = 7.30 \text{ g}$ [1] <br/> Actual mass at 85.0% yield = $7.30 \times 0.850 = 6.20 \text{ g}$ [1]                                                                                                                                                 | 3     |
| 37(a) | B (Bromoethane + ethanolic KCN) [1]                                                                                                                                                                                                                                                                                                                                                                     | 1     |
| 37(b) | Propanenitrile [1]                                                                                                                                                                                                                                                                                                                                                                                      | 1     |
| 38(a) | $\text{CH}_3\text{CH}(\text{OH})\text{CN} + 4[\text{H}] \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{NH}_2$ [1]                                                                                                                                                                                                                                                                          | 1     |
| 38(b) | Reduction occurs exclusively at the CN triple bond [1] <br/> None of the four bonds attached to the asymmetric chiral carbon ( $\text{C}_2$ ) are cleaved or reformed, preserving the stereochemical configuration [1]                                                                                                                                                                                  | 2     |
| 38(c) | $\text{CH}_3-\text{CH}(\text{OH})-\text{CH}_2-\text{NH}_2$ [1]                                                                                                                                                                                                                                                                                                                                          | 1     |
| 39(a) | In propanenitrile, the ethyl group is bonded directly to carbon ( $\text{CH}_3\text{CH}_2-\text{CN}$ ) [1] <br/> In ethyl isonitrile, the ethyl group is bonded directly to nitrogen ( $\text{CH}_3\text{CH}_2-\text{NC}$ with coordinate bond) [1]                                                                                                                                                     | 2     |
| 39(b) | KCN is purely ionic (K and $:\text{CN}$ ), where carbon carries the negative charge and acts as the stronger nucleophilic centre [1] <br/> Ag-C in AgCN is predominantly covalent; the carbon lone pair is engaged with silver, leaving the nitrogen lone pair free to act as the nucleophile [1]                                                                                                       | 2     |
| 40(a) | $\text{CH}_3\text{CH}_2\text{COOH} + \text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2 \rightarrow \text{CH}_3\text{CH}_2\text{COO}^- \text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_3^+$ (propylammonium propanoate) [1] <br/> Neutralisation / acid-base proton transfer [1]                                                                                                                                   | 2     |
| 40(b) | $\text{CH}_3\text{CH}_2\text{COCl} + \text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CONHCH}_2\text{CH}_2\text{CH}_3 + \text{HCl}$ [1]                                                                                                                                                                                                                             | 1     |
| 40(c) | N-propylpropanamide [1]                                                                                                                                                                                                                                                                                                                                                                                 | 1     |
| 40(d) | Add aqueous $\text{CuSO}_4$ dropwise then in excess to each solution [1] <br/> Propylamine produces a pale blue precipitate that dissolves in excess to form a deep royal-blue solution [1] <br/> Propanamide is non-basic and does not coordinate to $\text{Cu}^{2+}$ ; solution remains unchanged pale blue [1]                                                                                       | 3     |
| 41(a) | Amine group (secondary amine) [1]                                                                                                                                                                                                                                                                                                                                                                       | 1     |
| 41(b) | Formula $\text{C}_3\text{H}_9\text{N}$ with two $^1\text{H}$ NMR signals (ratio 6:3): $(\text{CH}_3)_2\text{CH}-\text{NH}_2$ (isopropylamine / 1-methylethylamine) has two methyls (6H) and $\text{CH}-\text{NH}_2$ (3H) [2] <br/> Systematic name: Propan-2-amine (or 1-methylethylamine) [1]                                                                                                          | 3     |

| Q     | EXAMINER MARKING POINTS / INDICATIVE CONTENT                                                                                                                                                                                                                                                                                                                                            | MARKS |
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| 41(c) | $\text{CH}_3\text{CONHCH}(\text{CH}_3)_2$ (N-(1-methylethyl)ethanamide) [1]                                                                                                                                                                                                                                                                                                             | 1     |
| 42(a) | 2200 – 2260 $\text{cm}^{-1}$ (sharp and distinct) [1]                                                                                                                                                                                                                                                                                                                                   | 1     |
| 42(b) | Primary amines have two N-H bonds ( $-\text{NH}_2$ ), allowing both symmetric and asymmetric N-H stretching vibrations, producing two separate absorption peaks [1]<br>Secondary amines have only one N-H bond, giving only a single N-H stretching absorption [1]                                                                                                                      | 2     |
| 42(c) | Tertiary amines ( $\text{R}_3\text{N}$ ) possess no N-H bonds, so no N-H stretching vibration is possible [1]                                                                                                                                                                                                                                                                           | 1     |
| 43(a) | $[\text{CH}_2=\text{NH}_2]^+ \leftrightarrow [\text{CH}_2=\text{NH}]^+$ (iminium ion, mass = 14 + 16 = 30) [2]                                                                                                                                                                                                                                                                          | 2     |
| 43(b) | Ethyl radical, $\cdot\text{CH}_2\text{CH}_3$ (mass = 29) [1]                                                                                                                                                                                                                                                                                                                            | 1     |
| 43(c) | In propan-2-amine, alpha-cleavage loses a methyl radical ( $\cdot\text{CH}_3$ , mass 15) [1]<br>The resulting iminium fragment is $[\text{CH}_3\text{CH}=\text{NH}_2]^+$ with $m/z = 59 - 15 = 44$ [1]                                                                                                                                                                                  | 2     |
| 44(a) | Step 1: Ethene + $\text{HBr} \rightarrow$ Bromoethane (electrophilic addition) [1]<br>Step 2: Bromoethane + excess ethanolic $\text{NH}_3$ , heat in sealed tube $\rightarrow$ Ethylamine [2]                                                                                                                                                                                           | 3     |
| 44(b) | Step 1: Bromoethane + $\text{KCN}$ in aqueous ethanol $\rightarrow$ Propanenitrile [1]<br>Step 2: Propanenitrile + dilute $\text{HCl}$ , reflux $\rightarrow$ Propanoic acid [1]<br>Step 3: Propanoic acid + $\text{SOCl}_2$ (or $\text{PCl}_5$ ) $\rightarrow$ Propanoyl chloride [1]                                                                                                  | 3     |
| 44(c) | $\text{CH}_3\text{CH}_2\text{COCl} + \text{CH}_3\text{CH}_2\text{NH}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CONHCH}_2\text{CH}_3 + \text{HCl}$ [1]<br>Room temperature / anhydrous conditions [1]                                                                                                                                                                                    | 2     |
| 45(a) | B (2-methylbutanoic acid) [1]                                                                                                                                                                                                                                                                                                                                                           | 1     |
| 45(b) | Correct skeletal formula of $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{COOH}$ [1]                                                                                                                                                                                                                                                                                               | 1     |
| 46(a) | Add aqueous copper(II) sulfate [1]<br>Propylamine forms a deep royal-blue solution / complex; propanamide gives no reaction / remains pale blue [1]                                                                                                                                                                                                                                     | 2     |
| 46(b) | $\text{LiAlH}_4$ in dry ether (or $\text{H}_2$ with heated $\text{Ni}$ catalyst) [2]                                                                                                                                                                                                                                                                                                    | 2     |
| 46(c) | Dilute hydrochloric acid (or dilute $\text{H}_2\text{SO}_4$ ), heat under reflux [2]                                                                                                                                                                                                                                                                                                    | 2     |
| 47(a) | Nucleophilic addition-elimination (condensation) [1]                                                                                                                                                                                                                                                                                                                                    | 1     |
| 47(b) | $\text{CH}_3\text{CH}_2\text{CH}=\text{NH} + \text{H}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ [1]                                                                                                                                                                                                                                                                   | 1     |
| 47(c) | $\text{CH}_3\text{-CH}(\text{NH}_2)\text{-CH}_3$ (propan-2-amine) [1]                                                                                                                                                                                                                                                                                                                   | 1     |
| 48(a) | B (Bromine water) [1]                                                                                                                                                                                                                                                                                                                                                                   | 1     |
| 48(b) | 2,4,6-tribromophenylamine [1]                                                                                                                                                                                                                                                                                                                                                           | 1     |
| 49(a) | Moles of $\text{HCl} = 0.01840 \times 0.150 = 2.76 \times 10^{-3} \text{ mol}$ [1]<br>Concentration of ethylamine = $(2.76 \times 10^{-3}) / 0.0200 = 0.138 \text{ mol dm}^{-3}$ [1]                                                                                                                                                                                                    | 2     |
| 49(b) | Acidic ( $\text{pH} < 7$ , typically $\text{pH} \sim 5\text{--}6$ ) [1]<br>The ethylammonium cation ( $\text{CH}_3\text{CH}_2\text{NH}_3^+$ ) is a weak Bronsted-Lowry acid that hydrolyses in water to release $\text{H}_3\text{O}^+$ ions: $\text{CH}_3\text{CH}_2\text{NH}_3^+ + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{CH}_2\text{NH}_2 + \text{H}_3\text{O}^+$ [1] | 2     |
| 49(c) | Methyl orange [1]<br>For a weak base-strong acid titration, the vertical $\text{pH}$ drop occurs in the acidic region ( $\text{pH} \sim 3.5\text{--}6.5$ ), which matches the colour change interval of methyl orange [1]                                                                                                                                                               | 2     |
| 50(a) | Ammonia gas ( $\text{NH}_3$ ) OR methylamine ( $\text{CH}_3\text{NH}_2$ ) [1]                                                                                                                                                                                                                                                                                                           | 1     |
| 50(b) | Carboxylate is propanoate (3 carbons) and total carbons = 4 $\rightarrow$ amide must have N-methyl group: $\text{CH}_3\text{CH}_2\text{CONHCH}_3$ (N-methylpropanamide) OR if gas is methylamine [2]<br>Systematic name: N-methylpropanamide [1]                                                                                                                                        | 3     |

| Q     | EXAMINER MARKING POINTS / INDICATIVE CONTENT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | MARKS |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 50(c) | Formula of W: $\text{CH}_3\text{CH}_2\text{CH}_2\text{NHCH}_3$ (secondary amine) OR if Z is butanamide ( $\text{CH}_3\text{CH}_2\text{CH}_2\text{CONH}_2$ , gas is $\text{NH}_3$ , carboxylate is butanoate): if propanoate is obtained, Z must be N-methylpropanamide; reduction yields N-methylpropan-1-amine (secondary). If W is a primary amine, Z must be butanamide: $\text{CH}_3\text{CH}_2\text{CH}_2\text{CONH}_2$ (hydrolysis yields butanoic acid). For acid = propanoic: Z has 3 C on acid side + 1 C on amine side $\rightarrow \text{CH}_3\text{CH}_2\text{CONHCH}_3$ [1]<br>Primary amine W:<br>Butan-1-amine: $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$ (if Z is butanamide) [1] | 2     |